Transformations of Conics and the Erlangen View of Geometry

N-20250211

§1.1 The organizing question

This chapter studies conic sections through transformations. The guiding question is not only how an ellipse, parabola, or hyperbola is described by an equation, but what kind of geometry regards two such objects as the same.

For transformations of the *same real affine plane*, the first part of a useful hierarchy really is

$$\text{Euclidean} \subset \text{similarity} \subset \text{affine} \subset \text{projective}.$$

Conformal and inversive geometry take another direction. A shear is affine and projective but does not preserve angles; inversion in a circle preserves angles (up to orientation) but generally takes straight lines to circles. The two families are therefore not successive levels of the same inclusion chain.

The interesting question is what each transformation actually preserves. This is the spirit of Klein's Erlangen program: a geometry concerns properties invariant under a specified group of transformations. We will test that principle with concrete conics, a cross-ratio calculation and an honest example of inversion.[1]

§1.2 Affine transformations

Definition 1.2.1. An affine transformation of $\mathbb{R}^n$ is a map

$$T(x) = Ax + b,$$

where A is a linear map and $b \in \mathbb{R}^n$. If $A \in GL(n, \mathbb{R})$, then T is an affine automorphism.

Affine transformations preserve straight lines, parallelism, ratios along a line, convexity, and barycentric combinations. They do not preserve lengths, angles, or circles.

Example 1.2.2

An ellipse can be sent to a circle by an affine transformation. If

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

then the affine map

$$(x, y) \mapsto (x/a, y/b)$$

sends it to the unit circle.

[1] For real projective conics, cross-ratios and polarity see Bobenko's *Geometry I* lecture notes, Sections 2.5 and 2.10–2.13.

§1.3 Conics in matrix form

A plane conic may be written as

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

In homogeneous coordinates $[x : y : z]$, this becomes

$$Ax^2 + Bxy + Cy^2 + Dxz + Eyz + Fz^2 = 0.$$

Equivalently,

$$X^T Q X = 0, \qquad X = (x, y, z)^T,$$

where Q is a symmetric 3×3 matrix.

Under a projective transformation $X \mapsto MX$, the conic transforms by

$$Q \mapsto M^{-T} Q M^{-1}.$$

Thus projective classification of conics becomes a question about symmetric bilinear forms up to change of coordinates.

A concrete matrix is the best antidote to the vague sentence that all nondegenerate conics are projectively equivalent. The parabola

$$C_1 : \quad y = x^2$$

has homogeneous equation

$$X^2 - YZ = 0,$$

so it is represented by

$$A_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} \\ 0 & -\frac{1}{2} & 0 \end{pmatrix}.$$

The unit circle

$$C_2 : \quad X^2 + Y^2 = Z^2$$

is represented by

$$A_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Now take

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix}.$$

A direct multiplication gives

$$M^T A_1 M = A_2.$$

Indeed, if $U = (u, v, w)^T$ and $X = MU = (u, v + w, -v + w)^T$, then

$$X^T A_1 X = u^2 - (v + w)(-v + w) = u^2 + v^2 - w^2 = U^T A_2 U.$$

Thus the projective change of coordinates $U = M^{-1}X$ sends the parabola equation to the circle equation.

This is not an affine transformation. The inverse matrix is

$$M^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

The unique real point at infinity of the parabola,

$$[0 : 1 : 0],$$

is sent to

$$M^{-1}[0 : 1 : 0] = [0 : 1 : 1],$$

a finite real point on the circle. Meanwhile the usual circular points at infinity of the real circle are the complex points

$$[1 : i : 0], \qquad [1 : -i : 0].$$

So the projective transformation has moved the affine line at infinity itself; this is exactly why a parabola and a circle can be the same projective conic while remaining different in affine geometry. The difference between them is absolute for affine geometry, but only a coordinate choice for projective geometry.

§1.4 Projective geometry

Projective geometry adds points at infinity. In the real projective plane,

$$\mathbb{P}^2(\mathbb{R}) = \{[x : y : z] \mid (x, y, z) \neq 0\}/\sim,$$

where

$$[x : y : z] = [\lambda x : \lambda y : \lambda z], \qquad \lambda \neq 0.$$

The affine plane is the chart $z \neq 0$, and the line at infinity is $z = 0$.

Definition 1.4.1. A projective transformation of $\mathbb{P}^2$ is induced by an invertible matrix

$$M \in GL(3, \mathbb{R}),$$

acting by

$$[X] \mapsto [MX].$$

Scalar multiples of M induce the same projective transformation. Thus the transformation group is

$$PGL(3, \mathbb{R}).$$

Projective transformations preserve incidence and cross-ratio, but not parallelism, distance, or angle. In projective geometry, all nondegenerate conics over an algebraically closed field are projectively equivalent.

§1.5 Cross-ratio

For four collinear points A, B, C, D, the cross-ratio is denoted

$$(A, B; C, D).$$

In an affine coordinate it has the form

$$(a, b; c, d) = \frac{(c-a)(d-b)}{(c-b)(d-a)}.$$

Theorem 1.5.1

A projective transformation of a projective line preserves the cross-ratio of four ordered distinct points. The restriction of a projective transformation of $\mathbb{P}^2$ to any line has the same property.

Proof. In an affine chart, a projective change of line coordinate has the form

$$F(t) = \frac{\alpha t + \beta}{\gamma t + \delta}, \qquad \alpha\delta - \beta\gamma \neq 0.$$

For any two finite coordinates where the denominators do not vanish,

$$F(s) - F(t) = \frac{(\alpha\delta - \beta\gamma)(s-t)}{(\gamma s + \delta)(\gamma t + \delta)}.$$

The cross-ratio is a quotient of four such differences. Each factor $\alpha\delta - \beta\gamma$ and each denominator $\gamma t + \delta$ appears equally often in its numerator and denominator, and cancels. Hence

$$(F(a), F(b); F(c), F(d)) = (a, b; c, d).$$

Points at infinity are handled by taking limits in projective coordinates. For a transformation of a projective plane, restrict its defining linear map to the two-dimensional vector space representing the given line; the same calculation applies. □

For an actual numerical example, take $(a, b, c, d) = (0, 1, 2, 3)$. Then

$$(0, 1; 2, 3) = \frac{(2-0)(3-1)}{(2-1)(3-0)} = \frac{4}{3}.$$

The non-affine projective map $F(t) = t/(t+1)$ sends these four positions to $0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}$, whose cross-ratio is

$$\frac{(\frac{2}{3}-0)(\frac{3}{4}-\frac{1}{2})}{(\frac{2}{3}-\frac{1}{2})(\frac{3}{4}-0)} = \frac{4}{3}.$$

The individual interval lengths have changed. The special combination has not. Since a projectivity of a line is determined by the images of three distinct points, the cross-ratio measures the position of a fourth point relative to those three in a way that survives perspective.

Remark 1.5.2 — The cross-ratio is the prototype of a projective invariant. Once one understands it, projective geometry stops looking like distorted Euclidean geometry and becomes an invariant theory of incidence.

§1.6 Inversion: when a straight line turns into a circle

Inversion about the unit circle is defined on $\mathbb{R}^2 \setminus \{0\}$ by

$$I(x) = \frac{x}{\|x\|^2}.$$

It is an involution: $I(I(x)) = x$. The unit circle remains pointwise fixed; points inside it exchange places with points outside. In complex notation $I(z) = 1/\overline{z}$.

Take the line $x = c$, $c \neq 0$, and write the coordinates of its inverted image as (u, v). Since inversion is its own inverse,

$$\frac{u}{u^2 + v^2} = c \quad \Longleftrightarrow \quad \boxed{\left(u - \frac{1}{2c}\right)^2 + v^2 = \frac{1}{4c^2}.}$$

The image is a circle of radius $1/(2|c|)$ centered at $(1/(2c), 0)$, with the origin omitted. The origin is its limiting point corresponding to points running infinitely far out along the original line. In the other direction, circles passing through the center of inversion become lines.

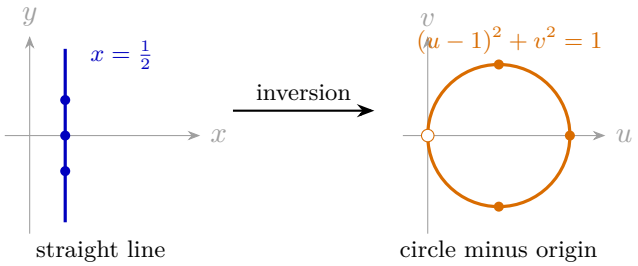


Figure 1.1: Under unit-circle inversion, $(\frac{1}{2}, 0)$ maps to $(2, 0)$ and $(\frac{1}{2}, \pm\frac{1}{2})$ map to $(1, \pm 1)$. The hollow origin is excluded, though the image circle passes through it as a limiting point.

But inversion has an entirely different invariant from projective maps. Put $u = x/\|x\|$. Differentiation gives

$$DI_x = \frac{1}{\|x\|^2}(I_2 - 2uu^T).$$

The matrix $I_2 - 2uu^T$ is a reflection: its square is the identity, it has eigenvalues $1, -1$, and it preserves inner products. Hence the derivative of inversion is a positive scale factor followed by an orthogonal reflection. It preserves *unoriented* angles locally, while reversing orientation. In complex geometry it is anti-conformal (anti-holomorphic), rather than an orientation-preserving holomorphic map.

This already refutes the proposed inclusion of all projective maps inside the conformal group, or vice versa. The affine shear $(x, y) \mapsto (x + y, y)$ is projective but changes a right angle. Inversion preserves angles but maps our straight line to a genuine circle, and so cannot be a projective plane transformation, which must send lines to lines.

§1.7 Möbius transformations

A Möbius transformation has the form

$$z \mapsto \frac{az + b}{cz + d}, \qquad ad - bc \neq 0.$$

It acts naturally on the Riemann sphere

$$\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}.$$

The group of Möbius transformations is

$$PGL(2, \mathbb{C}).$$

Möbius transformations send generalized circles to generalized circles and preserve the *complex* cross-ratio. They are projective transformations of the complex projective line $\mathbb{P}^1(\mathbb{C})$ and precisely the orientation-preserving conformal automorphisms of the Riemann sphere. This is a projective space of complex dimension one, not the real projective plane $\mathbb{P}^2(\mathbb{R})$ used for ordinary conics.

Circle inversion $z \mapsto 1/\overline{z}$, on the other hand, is anti-Möbius: conjugation reverses orientation and conjugates the complex cross-ratio. It preserves the underlying real-circle geometry without belonging to the holomorphic Möbius group. These distinctions explain how projective and conformal structures can meet in one complex dimension while remaining incomparable as transformation groups of the Euclidean plane.

§1.8 A hierarchy of invariants

Transformation group	Preserved	Forgotten
Euclidean	Distance, angle, incidence	Coordinates
Similarity	Angle, ratios of lengths	Absolute scale
Affine	Incidence, parallelism, ratios on a line	Angle and length
Projective	Incidence, cross-ratio	Parallelism and affine ratio
Möbius	Generalized circles, angles, cross-ratio on $\widehat{\mathbb{C}}$	Euclidean size

§1.9 Conics and the Line at Infinity

Fix the affine chart $Z = 1$, with line at infinity $Z = 0$. A real affine ellipse, parabola and hyperbola can have projective completions

$$\begin{aligned} X^2 + Y^2 - Z^2 &= 0 \quad \text{ellipse (unit circle),} \\ X^2 - YZ &= 0 \quad \text{parabola,} \\ X^2 - Y^2 - Z^2 &= 0 \quad \text{hyperbola.} \end{aligned}$$

Their intersection with the *fixed* line $Z = 0$ gives the three calculations

	equation at infinity	real intersections
ellipse	$X^2 + Y^2 = 0$	none
parabola	$X^2 = 0$	one double point
hyperbola	$X^2 - Y^2 = 0$	two distinct points

A projective parabola is tangent to the line at infinity, while a projective hyperbola meets it in the two directions of its asymptotes. The circle has two complex points at infinity $[1 : \pm i : 0]$, and no real ones. The intersection pattern is preserved under affine transformations because the line at infinity stays where it is; a general projective map may move this line.

Over the reals one must also be careful with the claim that all nonsingular conics are projectively equivalent. A nondegenerate symmetric quadratic form in three variables has a signature: the numbers of positive and negative squares after a real invertible change of coordinates. Sylvester's law of inertia says that signature is preserved. If the form is definite, the equation $X^TQX = 0$ has no nonzero real solution, and its projective conic is empty. If it is indefinite, multiplying its equation by -1 when necessary and changing coordinates brings it to

$$U^2 + V^2 - W^2 = 0.$$

Thus all *nonempty nonsingular real* projective conics are equivalent; an empty definite conic forms a separate case. Over an algebraically closed field of characteristic different from 2, nonsingular plane conics do form one projective equivalence class.

This explains precisely what was achieved in the parabola-to-circle matrix calculation: we changed the chosen affine chart while staying within the real indefinite projective class. We did not claim an arbitrary real quadratic form can be changed into an ellipse.

§1.10 Polar lines

For a nonsingular conic $X^TQX = 0$, a point P has a polar line

$$P^TQX = 0.$$

If P lies on the conic, the polar line is the tangent line at P. If P lies outside, the polar encodes the chord of contact of tangents from P.

The pole-polar relation is one of the best examples of why projective conic theory is richer than coordinate classification. It converts points to lines and lines to points in a way controlled by the conic.

The definition actually comes from differentiation. If $F(X) = X^TQX$ with symmetric Q, then

$$DF_P(V) = 2P^TQV.$$

At a point P on the conic, the projective tangent line is therefore the equation $P^TQX = 0$. Since Q is invertible, the assignment from a point P to the covector QP is itself an isomorphism (up to scale). Thus a conic establishes a precise duality between points and lines.

Here is a calculation that makes the polar line visible. For the unit circle $X^2 + Y^2 - Z^2 = 0$, choose the exterior point

$$P = [2 : 0 : 1], \qquad Q = \operatorname{diag}(1, 1, -1).$$

Its polar is $2X - Z = 0$, or $x = 1/2$ in the affine plane. The polar meets the circle at

$$T_{\pm} = \left(\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\right).$$

The tangent at $T_{\pm}$ has equation

$$\frac{1}{2}x \pm \frac{\sqrt{3}}{2}y = 1.$$

Both tangent lines pass through $(2, 0)$: substituting $x = 2$, $y = 0$ makes either left-hand side equal to 1. Hence the polar of this exterior point is exactly the chord of contact

of its two tangents. This picture is an incidence statement controlled entirely by the bilinear form.

The reciprocity of polarity is just the symmetry of Q. If R belongs to the polar line of P, then $P^TQR = 0$. Symmetry gives $R^TQP = 0$, so P belongs to the polar line of R. What sounds like a rather mystical point–line duality is a one-line identity in linear algebra.

§1.11 Parametrizing a circle by lines through one point

The same circle connects projective geometry to rational functions. Fix the point $(0, -1)$ on $x^2 + y^2 = 1$, and pass the line $y = tx - 1$ through it. Substitution gives

$$x^2 + (tx - 1)^2 = 1 \quad \Longleftrightarrow \quad x((1 + t^2)x - 2t) = 0.$$

The root $x = 0$ is the fixed base point; the second intersection is

$$\boxed{x = \frac{2t}{1 + t^2}, \qquad y = \frac{t^2 - 1}{1 + t^2}.}$$

The missing top point $(0, 1)$ is reached at the projective parameter $t = \infty$. In homogeneous coordinates the parametrization is

$$[S : T] \longmapsto [2ST : T^2 - S^2 : S^2 + T^2].$$

We have exhibited a rational parametrization of a nonsingular conic by the projective line of slopes through one point. The half-angle formulas of trigonometry conceal the same idea: $\tan(\theta/2)$ turns points on a circle into rational coordinates.

§1.12 How to choose the right geometry

The Erlangen viewpoint becomes useful only when it tells us which tools are allowed. For conic problems, a practical guide is:

Question being asked	Natural geometry	Reason
length, angle, focus, eccentricity	Euclidean/similarity	metric data is essential
midpoints, parallelism, area ratios	affine	affine maps preserve barycentric structure
incidence, tangency, conic equivalence	projective	the line at infinity may move
cross-ratio or four points on a line	projective	cross-ratio is projectively invariant
angle with generalized circles	conformal/inversive	inversion preserves angles and generalized circles

Remark 1.12.1 (A common mistake) — Saying that two conics are projectively equivalent does not mean they are the same in Euclidean or affine geometry. A parabola and a circle may be projectively equivalent, but the projective map can move the line at infinity. Once that happens, affine notions such as parallelism and the distinction between ellipse/parabola/hyperbola need not be preserved.